MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3220 GRINDING AND CLASSIFICATION-BALL MILL EQUIPMENT FOUNDATION & MISC. FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of Equipment & Micslanious Foundation (Cyclone Overflow Pump Sump 3220-TNK-002, Conditioning Feed Pump 3220-PMP-006 & 007, Particle Size Analyser Sample Return Pump 3310-PMP-502, Cyclone Cluster Feed Pump 3220-PMP-001 & 002, Gravity Concentrator Feed Pump 3220-PMP-004 & 005, Ball Mill Discharge Sump 3220-TNK-001 & Retaining Wall for Mill Ball Bunker) at Ball Mill Area (3220) Grade Slab.
The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in standard excel sheet. Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet. For strength calculation Bending moment & Shear Force values are calculate in the conventional method & designed in the standard excel sheet.

4. DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A02 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.A.(i) Density of Dry Soil : 1.9 T/m^3
4.2.A.(ii) Density of Concrete : 2.5 T/m^3
4.2.A.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 50 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	MinimumCove (mm)
Exposed to Soil	Foundation	50
	Wall	50
	Column / Pedestal	40

5. LOAD

5.1 Primary Loads

For equipment foundation design the primary load values are taken from GRINDING AND CLASSIFICATION - BALL MILL LOAD TABLE Drg. No. : MD-216-3220-EG-ME-GAD-1089

1 DEAD LOAD (D)
2 OPERATING LOAD (OL)

5.2 Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
101	DL+OL

5.3 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
201	1.4(DL+OL)

6. SAFETY FACTOR AGAINST STABILITY

6.1 Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0
6.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7. ALLOWABLE SOIL BEARING CAPACITY

All the equipment supporting pedestals are placed over Grade Slab type **Heavy Duty Paving**. As per Standard Detail of Paving (MD-216-1000-EG-CV-DED-1002) below the equipment supporting pedestal Net Safe Bearing Capacity maintain within **50 KN/m2**. Only the Heavy weight equipment Grade Slab below pedestal additionally thicken 100mm and maintain the Net Safe Bearing Capacity within **50 KN/m2**.

For Retaining Wall design we have considered the Net Safe Bearing Capacity at founding level as **200 KN/m2**.

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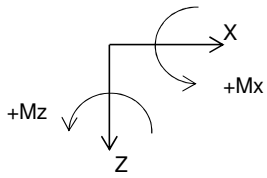
8.1 DESIGN OF FOUNDATION PEDESTAL MKD. P1 :

DESIGN DATA :

Net Safe Bearing Capacity =	5	Mton/m^2	*(Ref. Drg. No. : MD-216-1000-EG-CV-DED-1002)*
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	

FOUNDATION DIMENSION :

Length of Column (lx) =	0.5	m
Width of Column (lz) =	0.5	m
Ht. of the Column =	0.3	m
No. of the Column =	3	
Thk. of Base Slab =	0.25	m
Length of Foundation (Lx) =	5	m
Width of Foundation (Lz) =	5	m



WT. OF FOUNDATION (Wf) :

ITEM	WT. CALCULATION (Mton)	
Column =	0.5X0.5X0.3X3X2.5 =	0.563
	Total =	0.563

LOAD FOR STRESS CALCULATION :

(Ref. Drg. No. : MD-216-3220-EG-GE-GAD-1076-A01)

DEAD LOAD (D) =	58.1	Mton
OPERATING LOAD (OL) =	0	Mton

BASE PRESSURE CHECK:

Max. Base Pressure = (58.1+0+0.563)/5/5
= 2.347 Mton/m^2 **< 5Mton/m2 , O.K.**

REINFORCEMENT & PAVING TYPE

Provided **250** mm Thk. Heavt Duty Paving below Pedestal
Provided **D12** Welded Wire Mesh @ 200 X 200 - Double Layer

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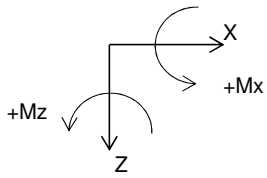
8.2 DESIGN OF FOUNDATION PEDESTAL MKD. P2 & P3 :

DESIGN DATA :

Net Safe Bearing Capacity =	5	Mton/m^2	*(Ref. Drg. No. : MD-216-1000-EG-CV-DED-1002)*
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	

FOUNDATION DIMENSION :

Length of Column (lx) =	2.4	m
Width of Column (lz) =	1.55	m
Ht. of the Column =	0.3	m
No. of the Column =	1	
Thk. of Base Slab =	0.15	m
Length of Foundation (Lx) =	2.7	m
Width of Foundation (Lz) =	1.85	m



WT. OF FOUNDATION (Wf) :

ITEM	WT. CALCULATION (Mton)	
Column =	2.4X1.55X0.3X1X2.5 =	2.79
	Total =	2.79

LOAD FOR STRESS CALCULATION :

(Ref. Drg. No. : MD-216-3220-EG-ME-GAD-1089-A03)

DEAD LOAD (D) =	4.9	Mton
OPERATING LOAD (OL) =	7.3	Mton

BASE PRESSURE CHECK:

Max. Base Pressure = (4.9+7.3+2.79)/2.7/1.85

= 3.001 Mton/m^2 **< 5Mton/m2 , O.K.**

REINFORCEMENT & PAVING TYPE

Provided	**150**	mm Thk. Heavt Duty Paving below Pedestal
Provided	**D8**	Welded Wire Mesh @ 200 X 200 - Double Layer

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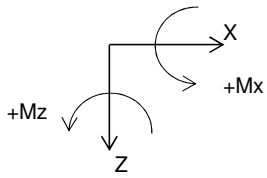
8.3 DESIGN OF FOUNDATION PEDESTAL MKD. P4 :

DESIGN DATA :

Net Safe Bearing Capacity =	5	Mton/m^2	*(Ref. Drg. No. : MD-216-1000-EG-CV-DED-1002)*
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	

FOUNDATION DIMENSION :

Length of Column (lx) =	1.65	m
Width of Column (lz) =	1.4	m
Ht. of the Column =	0.803	m
No. of the Column =	1	
Thk. of Base Slab =	0.15	m
Length of Foundation (Lx) =	1.95	m
Width of Foundation (Lz) =	1.7	m



WT. OF FOUNDATION (Wf) :

ITEM	WT. CALCULATION (Mton)	
Column =	1.65X1.4X0.803X1X2.5 =	4.637
	Total =	4.637

LOAD FOR STRESS CALCULATION :

(Ref. Drg. No. : MD-216-3220-EG-ME-GAD-1089-A03)

DEAD LOAD (D) =	3.1	Mton
OPERATING LOAD (OL) =	4.7	Mton

BASE PRESSURE CHECK:

Max. Base Pressure = (3.1+4.7+4.637)/1.95/1.7

= 3.752 Mton/m^2 **< 5Mton/m2 , O.K.**

REINFORCEMENT & PAVING TYPE

Provided	**150**	mm Thk. Heavt Duty Paving below Pedestal
Provided	**D8**	Welded Wire Mesh @ 200 X 200 - Double Layer

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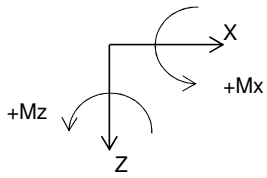
8.4 DESIGN OF FOUNDATION PEDESTAL MKD. P5 & P6 :

DESIGN DATA :

Net Safe Bearing Capacity =	5	Mton/m^2	*(Ref. Drg. No. : MD-216-1000-EG-CV-DED-1002)*
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	

FOUNDATION DIMENSION :

Length of Column (lx) =	5.7	m
Width of Column (lz) =	1.75	m
Ht. of the Column =	0.729	m
No. of the Column =	1	
Thk. of Base Slab =	0.25	m
Length of Foundation (Lx) =	6.7	m
Width of Foundation (Lz) =	2.75	m



WT. OF FOUNDATION (Wf) :

ITEM	WT. CALCULATION (Mton)	
Column =	5.7X1.75X0.729X1X2.5 =	18.179
	Total =	18.179

LOAD FOR STRESS CALCULATION :

(Ref. Drg. No. : MD-216-3220-EG-ME-GAD-1089-A03)

DEAD LOAD (D) =	27.9	Mton
OPERATING LOAD (OL) =	41.8	Mton

BASE PRESSURE CHECK:

Max. Base Pressure = (27.9+41.8+18.179)/6.7/2.75

= 4.77 Mton/m^2 **< 5Mton/m2 , O.K.**

REINFORCEMENT & PAVING TYPE

Provided **250** mm Thk. Heavt Duty Paving below Pedestal

Provided **D10** Welded Wire Mesh @ 200 X 200 - Double Layer

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8.5 DESIGN OF FOUNDATION PEDESTAL MKD. P7 & P8 :

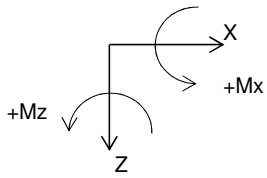
DESIGN DATA :

Net Safe Bearing Capacity = 5 Mton/m^2 *(Ref. Drg. No. : MD-216-1000-EG-CV-DED-1002)*

Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinf. , fy =	420	N/mm^2
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinf. , Φ =	0.9	
Strength Reduction Factor of Shear Reinf. , Φt =	0.75	
Clear Cover =	50	mm
Density of Concrete =	2.5	Mton/m^3
Density of Dry Soil =	1.9	Mton/m^3
Density of Saturated Soil =	2	Mton/m^3
Density of Water =	1	Mton/m^3

FOUNDATION DIMENSION :

Length of Column (lx) =	1.619	m
Width of Column (lz) =	1.373	m
Ht. of the Column =	0.803	m
No. of the Column =	1	
Thk. of Base Slab =	0.15	m
Length of Foundation (Lx) =	1.919	m
Width of Foundation (Lz) =	1.673	m



WT. OF FOUNDATION (Wf) :

ITEM	WT. CALCULATION (Mton)	
Column =	1.619X1.373X0.803X1X2.5 =	4.462
	Total =	4.462

LOAD FOR STRESS CALCULATION :

(Ref. Drg. No. : MD-216-3220-EG-ME-GAD-1089-A03)

DEAD LOAD (D) =	3.1	Mton
OPERATING LOAD (OL) =	4.7	Mton

BASE PRESSURE CHECK:

Max. Base Pressure = (3.1+4.7+4.462)/1.919/1.673
= 3.819 Mton/m^2 **< 5Mton/m2 , O.K.**

REINFORCEMENT & PAVING TYPE

Provided **150** mm Thk. Heavt Duty Paving below Pedestal
Provided **D8** Welded Wire Mesh @ 200 X 200 - Double Layer

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8.6 DESIGN OF FOUNDATION PEDESTAL MKD. P9.1 To P9.4 :

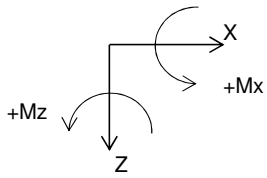
DESIGN DATA :

Net Safe Bearing Capacity = 5 Mton/m^2 *(Ref. Drg. No. : MD-216-1000-EG-CV-DED-1002)*

Compressive Strength of Concrete , fc' = 35 N/mm^2
Yield Strength of Main Reinf. , fy = 420 N/mm^2
Yield Strength of Shear Reinf. , fyt = 420 N/mm^2
Strength Reduction Factor of Main Reinf. , Φ = 0.9
Strength Reduction Factor of Shear Reinf. , Φt = 0.75
Clear Cover = 50 mm
Density of Concrete = 2.5 Mton/m^3
Density of Dry Soil = 1.9 Mton/m^3
Density of Saturated Soil = 2 Mton/m^3
Density of Water = 1 Mton/m^3

FOUNDATION DIMENSION :

Length of Column (lx) = 9.46 m
Width of Column (lz) = 1.9 m
Ht. of the Column = 0.7 m
No. of the Column = 1
Thk. of Base Slab = 0.25 m
Length of Foundation (Lx) = 9.96 m
Width of Foundation (Lz) = 6.295 m



WT. OF FOUNDATION (Wf) :

ITEM	WT. CALCULATION (Mton)	
Column =	9.46X1.9X0.7X1X2.5 =	31.455
	Total =	31.455

LOAD FOR STRESS CALCULATION :

(Ref. Drg. No. : MD-216-3220-EG-GE-GAD-1048-A01)

DEAD LOAD (D) = 257.9 Mton
OPERATING LOAD (OL) = 0 Mton

BASE PRESSURE CHECK:

Max. Base Pressure = (257.9+0+31.455)/9.96/6.295
= 4.615 Mton/m^2 **< 5Mton/m2 , O.K.**

REINFORCEMENT & PAVING TYPE

Provided **250** mm Thk. Heavt Duty Paving below Pedestal
Provided **D12** Welded Wire Mesh @ 200 X 200 - Double Layer

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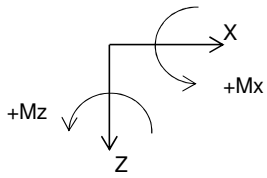
8.7 DESIGN OF FOUNDATION PEDESTAL MKD. P52 :

DESIGN DATA :

Net Safe Bearing Capacity =	5	Mton/m^2	*(Ref. Drg. No. : MD-216-1000-EG-CV-DED-1002)*
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	

FOUNDATION DIMENSION :

Length of Column (lx) =	1.1	m
Width of Column (lz) =	1.07	m
Ht. of the Column =	0.35	m
No. of the Column =	1	
Thk. of Base Slab =	0.15	m
Length of Foundation (Lx) =	1.4	m
Width of Foundation (Lz) =	1.37	m



WT. OF FOUNDATION (Wf) :

ITEM	WT. CALCULATION (Mton)	
Column =	1.1X1.07X0.35X1X2.5 =	1.03
	Total =	1.03

LOAD FOR STRESS CALCULATION :

(Ref. Drg. No. : MD-216-3220-EG-ME-GAD-1089-A03)

DEAD LOAD (D) =	0.8	Mton
OPERATING LOAD (OL) =	1.1	Mton

BASE PRESSURE CHECK:

Max. Base Pressure = (0.8+1.1+1.03)/1.4/1.37

= 1.528 Mton/m^2 **< 5Mton/m2 , O.K.**

REINFORCEMENT & PAVING TYPE

Provided	**150**	mm Thk. Heavt Duty Paving below Pedestal
Provided	**D8**	Welded Wire Mesh @ 200 X 200 - Double Layer

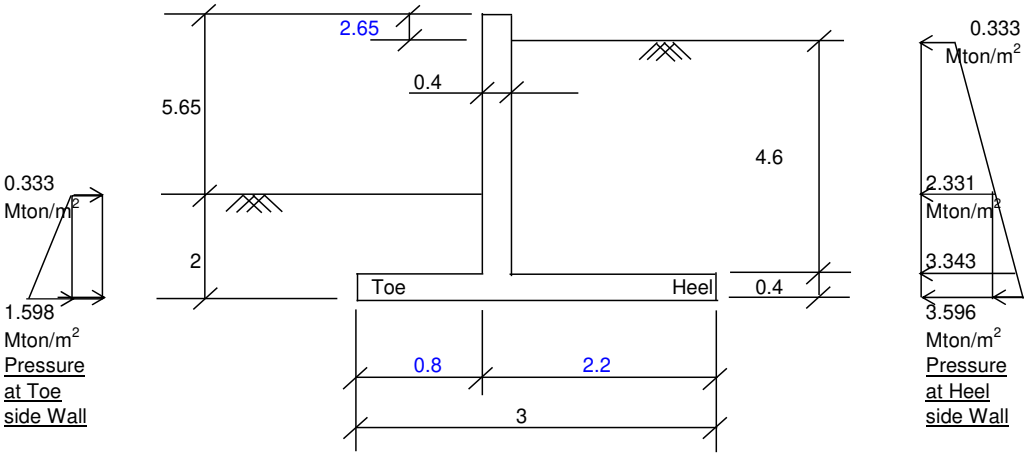
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9.1 Design of Retaining Wall for Mill Ball Bunker : (DRY CONDITION)



Design Data

Net SBC of Soil =	20	Mton/m^2	*(From BH-02)*
Gross SBC of Soil =	20+2X1.9 =	23.8 Mton/m^2	
Density of Retained Material =	2	Mton/m^3	*(Consider 1/4 area of the Ball contact with the Wall)*
Density of Dry Soil =	1.9	Mton/m^3	
Density of Concrete =	2.5	Mton/m^3	
Density of Water =	1	Mton/m^3	
Surcharge over Heel Slab =	1	Mton/m^2	*(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 5.9.3(i))*
Surcharge over Toe Slab =	1	Mton/m^2	*(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 5.9.3(i))*
Angle of Internal Friction (ϕ) =	30	Degree	
Active Pressure Coefficient (K_2) =	0.333		
Passive Pressure Coefficient =	3		

Dimension

Depth of Foundation =	2	m
Ht. of Retaining Wall above FGL =	5.65	m
Thickness of Retaining wall =	0.4	m
Thickness of Base Slab =	0.4	m
Consider, Length of Retaining Wall =	1	m

Load Calculation

Side Pressure over Retaining Wall at Heel Side

1. Due to Retained Material =	0.333X3X2 =	1.998	Mton/m^2
2. Due to Dry Soil =	0.333X2X1.9 =	1.265	Mton/m^2
3. Due to Surcharge =	0.333X1 =	0.333	Mton/m^2

Side Pressure over Retaining Wall at Toe Side

5. Due to Dry Soil =	0.333X2X1.9 =	1.265	Mton/m^2
6. Due to Surcharge =	0.333X1 =	0.333	Mton/m^2

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Check for Overturning

Calculation of Restoring Moment

Item Mkd.	Width (m)	Length (m)	Thk. / Ht. (m)	Density (Mton/m^3)	%	Wt. (T)	C.G. from Edge of Toe (m)	Clockwise Moment (Mton-m)
Heel Slab =	1.8	1	0.4	2.5	1	1.800	2.100	3.78
Toe Slab =	0.8	1	0.4	2.5	1	0.800	0.400	0.32
Retaining Wall =	7.25	1	0.4	2.5	1	7.250	1.000	7.25
Retained Material over Heel Slab =	1.8	1	3	2	0.9	9.720	2.100	20.412
Soil over Heel Slab =	1.8	1	1.6	2	0.9	5.184	2.100	10.886
Soil over Toe Slab =	0.8	1	1.6	2	0.9	2.304	0.400	0.922
					Total =	27.058	Total =	43.57

Calculation of Overturning Moment

Item Mkd.	Anti Clock Wise Moment (Mton-m)		Span	Total Moment (Mton-m)
Retained Material Pressure above FGL =	0.5X1.998X3X(2+3/3) =	8.991	1	8.991
Retained Material Pressure below FGL =	1.998X2^2/2 =	3.996	1	3.996
Due to Dry Soil =	0.5X1.265X2^2/3 =	0.843	1	0.843
Due to Surcharge =	0.333X(2+3)^2/2 =	4.163	1	4.163
	Total =	17.993	Total =	17.993

Total Wt. of Structure = 9.850 Mton

Seismic Force acting on wall = 0.28 Mton *(Ref. Seismic Load Calculation)*

Due to Seismic , Overturning Moment = 0.28 X (2+5.65)/2 = 1.071 Mton-m *(Anti Clock Wise)*

Total Overturning Moment = 17.993 + 1.071X0.75 = 18.796 Mton-m *(Anti Clock Wise)* *(Ref. SBC:301-2007, CL. No. 10.9.6)*

Total Restoring Moment = 43.57 Mton-m *(Clock Wise)*

F.O.S. against Overturning = 43.57/18.796 = 2.32 > 2 **OK**

(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.2)

Check for Sliding

Calculation of Horizontal Sliding Force

Item Mkd.	Horizontal Sliding Force (Mton)		Span	Total Sliding Force (Mton)
Retained Earth Pressure above FGL =	0.5X1.998X3 =	2.997	1	2.997
Retained Earth Pressure below FGL =	1.998X1.6 =	3.197	1	3.197
Due to Dry Soil =	0.5X1.265X1.6 =	1.012	1	1.012
Due to Dry Surcharge =	0.333X(1.6+3) =	1.532	1	1.532
			Total =	8.738

Total Horizontal Sliding Force = 8.738+0.28 = 9.018 Mton

Corresponding Vertical Load, V = 27.058 Mton

Horizontal Frictional coefficient, μ = 0.55 *(Ref. Doc. No. : MD-216-0000-PM-GM-RPT-0003, C01 BFS-Section 7B, CL. NO. 7B.6.4.2)*

Fricrtional Force = μ * V = 14.882 Mton

F.O.S. against Sliding = 14.882/9.018 = 1.65 > 1.5 **OK**

(Ref. Doc. No. : MD-101-SMEM-EG-ST-CRT-0001, C01, CL. No. 6.3.2)

Calculation of Base Pressure

C.G. of Load = 43.57/27.058 = 1.61 m

Eccentricity, e = 1.61-(0.8+2.2)/2 = 0.110 m

Moment due to Eccentricity = 0.11X27.058 = 2.976 Mton-m (Clockwise)

Overturning Moment = 17.993 Mton-m (Anti-Clockwise)

Net Moment = 17.993-2.976 = 15.017 Mton-m (Anti-Clockwise)

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Max. Base Pressure = (27.058/3/1)+(15.017X6/1/3^2) = 19.031 Mton/m^2 **< 23.8 Mton/m2, OK**
Min. Base Pressure = (27.058/3/1)-(15.017X6/1/3^2) = -0.992 Mton/m^2

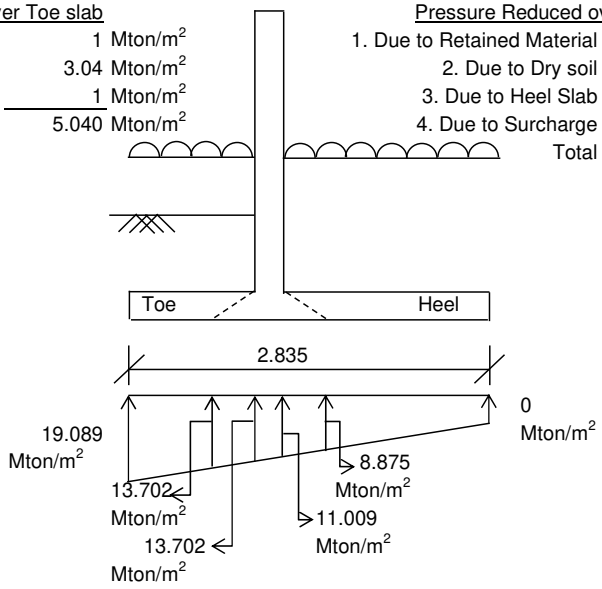
Modified Base Pressure

Max. Base Pressure = 19.089 Mton/m^2 **< 23.8 Mton/m2, OK**
Min. Base Pressure = 0 Mton/m^2

Base Pressure Diagram

Pressure Reduced over Toe slab
1. Due to Surcharge = 1 Mton/m^2
2.Due to Dry soil = 3.04 Mton/m^2
3. Due to Toe Slab = 1 Mton/m^2
Total = 5.040 Mton/m^2

Pressure Reduced over Heel slab
1. Due to Retained Material = 6 Mton/m^2
2. Due to Dry soil = 3.04 Mton/m^2
3. Due to Heel Slab = 1 Mton/m^2
4. Due to Surcharge = 1 Mton/m^2
Total = 11.040 Mton/m^2



Design of Heel slab

B.M. at Face of the Wall = -4.946 Mton/m^2
Mu = 4.946 X 1.4
= 6.924 Mton-m
Φ*Kn = Mu/bd^2 d eff. = 317 mm
= 6.924 X 10^7 / (1000 X 317^2)
= 0.689 N/mm^2 < 6.9 N/mm^2

Reqd. Pt = 0.0018
Min. Pt = 0.0018 *[SBC:304-CR-18, Table No. 7.6.1.1, Page-45]*
Reqd. , Ast = Pt*b*d.eff
= 0.0018X1000X317/10^2
= 5.71 cm^2

Design Parameter

Width =	1000	mm
Depth =	400	mm
Clear Cover =	75	mm
fc' =	35	N/mm^2
fy =	420	N/mm^2
fyt =	420	N/mm^2
Φ =	0.9	
Φt =	0.75	
Load Factor =	1.4	

Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Top
Provided **0** mm @ **200** C/C (Provide Ast = 0 cm2) at Top
Total Provide Ast = 10.05 cm2 at Top **O.K.**
Total Provide Pt = 0.0032

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SHEAR CHECK

S.F. at d-dist. from Wall Face = -8.702 Mton
Vu = 8.702X1.4
= 12.183 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X12.183X10^4X317/6.924/10^7)X1000X317/10^4
= 30.968 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^4
= 31.731 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X317/10^4
= 54.387 Mton
HENCE, Vc = 30.968 Mton
Φt*Vc = 0.75X30.968 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 23.226 Mton **> 12.183Mton , Hence O.K.**

Design of Heel slab

B.M. at Face of the Wall = 3.921 Mton/m^2
Mu = 3.921 X 1.4
= 5.489 Mton-m
Φ*Kn = Mu/bd^2 d eff. = 317 mm
= 5.489 X 10^7 / (1000 X 317^2)
= 0.546 N/mm^2 < 6.9 N/mm^2

Reqd. Pt = 0.0015
Min. Pt = 0.0018 *[SBC:304-CR-18, Table No. 7.6.1.1, Page-45]*
Reqd. , Ast = Pt*b*d.eff
= 0.0018X1000X317/10^2
= 5.71 cm^2

Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Bottom
Provided **0** mm @ **200** C/C (Provide Ast = 0 cm2) at Bottom
Total Provide Ast = 10.05 cm2) at Bottom **O.K.**
Total Provide Pt = 0.0032

SHEAR CHECK

S.F. at d-dist. from Wall Face = 5.485 Mton
Vu = 5.485X1.4
= 7.679 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X7.679X10^4X317/5.489/10^7)X1000X317/10^4
= 30.771 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^4
= 31.731 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X317/10^4
= 54.387 Mton
HENCE, Vc = 30.771 Mton
Φt*Vc = 0.75X30.771 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 23.078 Mton **> 7.679Mton , Hence O.K.**

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Design of Wall at Face of the Raft

B.M. at Face of the Raft = 14.305 Mton/m^2
Mu = 14.305 X 1.4
= 20.027 Mton-m
Φ*Kn = Mu/bd^2 d eff. = 317 mm
= 20.027 X 10^7 / (1000 X 317^2)
= 1.993 N/mm^2 < 6.9 N/mm^2

Reqd. Pt = 0.0055
Min. Pt = 0.0025 *[SBC:304-CR-18, CL. No. 11.6, Page-80]*
Reqd. , Ast = Pt*b*d.eff
= 0.0055X1000X317/10^2
= 17.44 cm^2

Provided 16 mm @ 200 C/C (Provide Ast = 10.05 cm2) at Bottom Alternately Placed
Provided 16 mm @ 200 C/C (Provide Ast = 10.05 cm2) at Bottom Alternately Placed
Total Provide Ast = 20.1 cm2) at Bottom Alternately Placed **O.K.**
Total Provide Pt = 0.0063

SHEAR CHECK

S.F. at Face of the Raft = 8.535 Mton
Vu = 8.535X1.4
= 11.949 Mton
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0063X11.949X10^4X317/20.027/10^7)X1000X317/10^4
= 30.648 Mton
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0063)X1000X317/10^4
= 33.401 Mton
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X35^0.5X1000X317/10^4
= 54.387 Mton
HENCE, Vc = 30.648 Mton
Φt*Vc = 0.75X30.648 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 22.986 Mton **> 11.949Mton , Hence O.K.**

Design of Wall at FGL

B.M. at FGL = 4.496 Mton/m^2
Mu = 4.496 X 1.4
= 6.294 Mton-m
Φ*Kn = Mu/bd^2 d eff. = 317 mm
= 6.294 X 10^7 / (1000 X 317^2)
= 0.626 N/mm^2 < 6.9 N/mm^2

Reqd. Pt = 0.0017
Min. Pt = 0.0025 *[SBC:304-CR-18, CL. No. 11.6, Page-80]*
Reqd. , Ast = Pt*b*d.eff
= 0.0025X1000X317/10^2
= 7.93 cm^2

Provided 16 mm @ 200 C/C (Provide Ast = 10.05 cm2) at Bottom **O.K.**
Total Provide Pt = 0.0032

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MD-216-3220-EG-CV-CAL-1160	Mansourah Massarah Gold Project	Page	OF
900707_GAU01_ZG01_CED01_32207_OU1 00671085_R2	CALCULATION GRINDING AND CLASSIFICATION-BALL MILL EQUIPMENT FOUNDATION & MISC. FOUNDATION	Date	08.11.2023
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SHEAR CHECK

S.F. at Face of the Raft = 3.996 Mton

Vu = 3.996X1.4

= 5.594 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0032X5.594X10^4X317/6.294/10^7)X1000X317/10^4

= 30.492 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0032)X1000X317/10^4

= 31.731 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X317/10^4

= 54.387 Mton

HENCE, Vc = 30.492 Mton

Φt*Vc = 0.75X30.492 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 22.869 Mton **> 5.594Mton , Hence O.K.**

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900707_GAU01_ZG01_CED01_32207_OU10 0671085_R22	CALCULATION GRINDING AND CLASSIFICATION-BALL MILL EQUIPMENT FOUNDATION & MISC. FOUNDATION	Date	08.11.2023
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9.2 Seismic Load Calculation for Retaining Wall : *(As per SBC:303-2007, CL No. 7.2.6)*

Input Data :

Height of Wall (H) = 7.65 m

Max. Surcharge at top = 1 T/m^2

Slope of wall back with respect to vertical (θ) = 0 Degree

Angle of slope of Filling (β) = 0 Degree

Density of Dry Soil (γ) = 1.9 $Mton/m^3$

Horizontal Ground Acceleration (k_h) = 0.02

Vertical Ground Acceleration (k_v) = $0.5*k_h$ = 0.01

Angle of Internal Friction (ϕ) = 30 Degree

Angle of Friction between Wall & Soil (δ) = $0.75*\phi$ = 22.5 Degree

Seismic Inertia Angle (Ψ) = $tan^{-1}(\frac{k_h}{1-k_v})$

= 1.157 Degree

Modified Slope of wall back (θ*) = θ+Ψ

= 1.157 Degree

Modified Inclination of Soil Surface (β*) = β+Ψ

= 1.157 Degree

From Coulomb Theory, K_A (β,θ) = $\frac{Cos^2(\phi-\theta)}{Cos^2\theta * Cos(\delta+\theta) * [1+\sqrt{\frac{Sin(\delta+\phi)*Sin(\phi+\beta)}{Cos(\delta+\theta)*Cos(\beta-\theta)}}]\text{^}2}$

= 0.2963

From Coulomb Theory, K_A (β*,θ*) = $\frac{Cos^2(\phi-\theta^*)}{Cos^2\theta^* * Cos(\delta+\theta^*) * [1+\sqrt{\frac{Sin(\delta+\phi)*Sin(\phi+\beta^*)}{Cos(\delta+\theta^*)*Cos(\beta^*-\theta^*)}}]\text{^}2}$

= 0.3007

Due to Earth Pressure

Max value of Dynamic Increment = 0.5X1.9X7.65X(0.3007-0.2963)*3/2

= 0.048

Active Thrust Due to Dynamic Increment = 2X7.65/3X0.048

= 0.245 Mton

Due to Earth Surcharge

Due to Surcharge max value of Dynamick Increment = 2X1XCOS(0)/COS(0)X(0.3007-0.2963)

= 0.0088

Active Thrust Due to Dynamic Increment = 0.5X7.65X0.0088

= 0.034 Mton

Total Active Thrust Due to Dynamic Increment = 0.245+0.034

= **0.279** Mton

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900707_GAU01_ZG01_CED01_32207_OU1 00671085_R2	CALCULATION GRINDING AND CLASSIFICATION-BALL MILL EQUIPMENT FOUNDATION & MISC. FOUNDATION	Date	08.11.2023
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ATTACHMENT - 1 : MECHANICAL INPUT DRAWING.

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DRAWING NUMBER: MD-216-3220-EG-ME-GAD-1085 REV: A03

PREVAILING WIND DIRECTION
PLANT NORTH
32° E
GEODETIC NORTH
25° W
MECCA
N
S
E
W

CYCLONE OVERFLOW TRASH SCREEN O/S BUNKER
CYCLONE OVERFLOW TRASH SCREEN O/S BUNKER
E= 264226.040
N= 2506224.040
P10
B.O.B(+) 913.300
1370
705
TOC EL. +912.670
LP
SLOPE 1:50
TOC EL. +913.000
HP
1300
P1.1
370
P2
549
1020
P1.2
P1.3
845
P3
600 x 600 TYP
3220-TNK-002
CYCLONE OVERFLOW PUMP SUMP
B.O.B(+) 913.350 TYP
2400
R 2020
TYP
120
95
3220-PMP-103
TRASH SCREEN AREA SUMP PUMP
3220-013 HOLD
3220-PMP-006
CONDITIONING FEED PUMP
3220-011 HOLD
3220-PMP-007
CONDITIONING FEED PUMP
3220-011 HOLD
2400 x 1550 TYP
B.O.B(+) 913.350 TYP
INSTALLATION CRANE LOAD TO BE CONSIDERED BY CIVIL FORKLIFT / BOBCAT FRONT AXLE LOAD 65 kN
3220-002 HOLD
3220-PMP-003
GRAVITY CONCENTRATOR FEED PUMP
3220-001 HOLD
3220-PMP-102
MILL DISCHARGE END SUMP PUMP
2850
500
P11
B.O.B(+) 913.650
TOC EL. +912.772
LP
RAMP
MILL BALL SCATS BUNKER
1650 x 1400 TYP
771
P4
830
B.O.B(+) 913.700
3220-TNK-001
BALL MILL DISCHARGE SUMP
5700 x 1750 TYP
680
CANAL
HELD BRG.
15324
FREE BRG.
TOC EL. +912.772
LP
1345
2530
4000
P9
P9.2
P9.3
P9.4
P5
SLOPE 1:50
3319
20
20
90
80
TOC EL. +913.000
HP
TOS EL. +916.200
E=264227.827
N=2506191.786
TOC EL. +913.000
HP
B.O.B(+) 913.729 TYP
FORKLIFT FRONT AXLE LOAD 65 kN
3220-003 HOLD
3220-PMP-001
CYCLONE CLUSTER FEED PUMP
3220-004 HOLD
3220-PMP-002
CYCLONE CLUSTER FEED PUMP
3220-005 HOLD
3220-PMP-004
GRAVITY CONCENTRATOR FEED PUMP
3220-006 HOLD
3220-PMP-005
GRAVITY CONCENTRATOR FEED PUMP
P6
49
3215
P7
614
782
P8
B.O.B(+) 913.810 TYP
600 x 600 TYP
B.O.B(+) 913.700 TYP
3220-PMP-101
MILL FEED END SUMP PUMP
3220-007 HOLD
RAMP
P12
450
B.O.B(+) 913.650
TOC EL. +912.772
LP
CANAL
TOC EL. +913.000
HP
3542
TOC EL. +913.000
HP
SLOPE 1:50
SLOPE 1:50
INSTALLATION CRANE LOAD TO BE CONSIDERED BY CIVIL FORKLIFT / BOBCAT FRONT AXLE LOAD 65 kN
3230-001 HOLD
3230-PMP-101
GRAVITY CONCENTRATION SPILLAGE SUMP PUMP
P13
765
TOC EL. +912.784
LP
CANAL
P14
TOC EL. +912.784
LP
FENCE
E= 264209.858
N= 2506152.847
B.O.B(+) 913.650
1325
1435
B.O.B(+) 913.650
2415
1540
P15
3550
SLOPE 1:50
CANAL
LP
TOC EL. +913.000
HP
3620-001 HOLD
3620-PMP-101
INTENSIVE CYANIDATION AREA SUMP PUMP
3620-PKG-001
INTENSIVE CYANIDATION ENDOR PACKAGE
3620-002 HOLD
TOC EL. +913.000
HP
TOC EL. +913.000
HP
3500
3500
P50
3125
3220-010 HOLD
3220-PKG-302
3220-MIL-001
GEARBOX LUBE UNIT
P48
P49
3125
3220-009 HOLD
3220-PKG-303
3220-MIL-001
GEARBOX LUBE UNIT
1600
3220-PKG-301
3220-MIL-001
MAIN BEARING LUBE UNIT
3220-008 HOLD
MILL BALL BUNKER
MILL BALL BUNKER
TOC EL. +913.000
TOC EL. +913.000
UP
3310-PMP-502
PARTICLE SIZE ANALYZER SAMPLE RETURN PUMP
3310-008 HOLD
1400
P52
A03
SLOPE 1:50
TOC EL. +913.000
HP
TOC EL. +913.000
LP
SPACE RESERVATION FOR LINERS
300 LINERS EACH WEIGHT 40 kN
FENCE
5400
5200
5200
6250
6250
6250
6250
5500
5500
58450
4950
4950
4950
5700
5700
5600
5600
7000
7000
7000
3220/J
3220/I
3220/H
3220/G
3220/F
3220/E
3220/D
3220/C
3220/B
3220/A
3220/1
3220/2
3220/3
3220/4
3220/5
3220/6
3220/7
3220/8
3220/9
3220/10
3220/11

NOTE:
LOAD TABLE REFER DRAWING NUMBER MD-216-3220-EG-ME-GAD-1089.

PLAN AT EL.(+)913.000

0 1 10m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
CHECKERED PLATE
GRATING
GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS: +Y, +Z, +Mz, +X, +My, +Mx

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m² GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN.

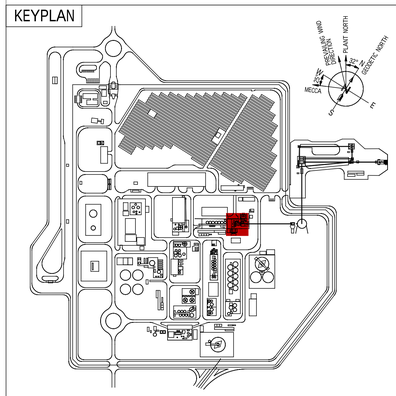
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3220-EG-ME-GAD-1066	GENERAL ARRANGEMENT DRAWING, PLAN AT EL(+) 913.000
MD-216-3220-EG-ME-GAD-1089	LOAD TABLE
MD-216-3220-EG-ME-GAD-1088	LOAD PLAN, PLAN AT EL (+)931.150
MD-216-3220-EG-ME-GAD-1087	LOAD PLAN, PLAN AT EL (+)924.550
MD-216-3220-EG-ME-GAD-1086	LOAD PLAN, PLAN AT EL (+)922.150

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A03	17-03-2020	REVISED AS MARKED	MB	TV	SC	RL	
A02	04-02-2020	ISSUED FOR DESIGN	SF	TV	SC	RL	
A01	03-01-2020	ISSUED FOR APPROVAL	BR	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
O190445_450_M_D_02372_001			2
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707_GAU01_ZG01_MLD14_32201		OU100667805	2
DRAWN	BR	03/01/2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	03/01/2020	LOAD PLAN
ENGINEER	SC	03/01/2020	GRINDING AND CLASSIFICATION - BALL MILL
APPROVED	RL	03/01/2020	PLAN AT EL.(+) 913.000
EMPLOYER			
SCALE 1:100	A0	DRAWING NUMBER: MD-216-3220-EG-ME-GAD-1085	REV A03

DRAWING NUMBER: MD-216-3220-EG-ME-GAD-1089 | REV: A03

LOADING POINT	P1.1*/.../1.3*				P2				P3				P4				P5				P6				P7				P8				P9.1/.../9.4 *				P10				P11				P12				P13				P14				P15				P16			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT					- 49				- 49				- 31				- 279				- 279				- 31				- 31								- 12				- 12				- 13				- 12				- 9				- 95				- 8			
2 DEAD LOAD (SECONDARY EQUIPMENT)																																																															
3 OPERATION LOAD 1. FILLING																																																															
4 OPERATION LOAD 2. PRESSURE																																																															
5 OPERATION LOAD 3. TEMPERATURE																																																															
6 OPERATION LOAD 4. DYNAMIC					- 73				- 73				- 47				- 418				- 418				- 47				- 47								- 18				- 17				- 19				- 17				- 13				- 11							
7 OPERATION LOAD 5. OTHERS																																																															
8 TEST LOAD FILLING / EMERGENCY																																																											- 400							
9 LIVE LOAD																																																											- 340							
10 WIND ±X - DIRECTION					-	2	-		-	2	-		-	1	-		-	1	-		-	1	-		-	1	-		-	1	-						-	-	-		-	-	-		-	-	-		-	-	-		-	-	-		-	10	-		-	1	-	
11 WIND ±Y - DIRECTION					-	-	1		-	-	1		-	-	1		-	-	6		-	-	6		-	-	1		-	-	1						-	-	-		-	-	-		-	-	-		-	-	-		-	-	-		-	-	32		-	-	1	
12 EARTHQUAKE ±X - DIRECTION					-	3	3		-	3	3		-	2	2		-	14	14		-	14	14		-	2	2		-	2	2						-	1	1		-	1	1		-	1	1		-	1	1		-	1	1		-	20	20		-	1	1	
13 EARTHQUAKE ±Y - DIRECTION					-	3	3		-	3	3		-	2	2		-	14	14		-	14	14		-	2	2		-	2	2						-	1	1		-	1	1		-	1	1		-	1	1		-	1	1		-	20	20		-	1	1	
EQUIPMENT - NO.	3220-TNK-002				3220-PMP-006				3220-PMP-007				3220-PMP-003				3220-PMP-001				3220-PMP-002				3220-PMP-004				3220-PMP-005				3220-TNK-001				3220-PMP-103				3220-PMP-102				3220-PMP-101				3230-PMP-101				3620-PMP-101				3620-PKG-001				3220-PMP-501			
DESCRIPTION	CYCLONE				CONDITIONING				CONDITIONING				GRAVITY				CYCLONE CUSTER				CYCLONE CUSTER				GRAVITY				GRAVITY				BALL MILL				TRASH SCREEN				MILL DISCHARGE				MILL FEED				GRAVITY				INTENSIVE				INTENSIVE				CYCLONE OVER.			
	OVERFLOW PUMP				FEED PUMP				FEED PUMP				CONCENTRATOR				FEED PUMP				FEED PUMP				CONCENTRATOR				CONCENTRATOR				DISCHARGE SUMP				AREA SUMP				END SUMP				END SUMP				SEPERATION				CYANIDATION				CYANIDATION				METALLURGICAL			
	SUMP												FEED PUMP												FEED PUMP				FEED PUMP								PUMP				PUMP				PUMP				SPILLAGE				AREA SUMP				VENDOR PACKAGE.				SAMPLE			
																																																									SUMP PUMP				PUMP								TRANSFER PUMP			

LOADING POINT	P17				P18				P19				P20.1/.../20.4*				P21.1/.../21.6				P22.1/.../22.4				P23.1/.../23.4				P24.1/.../24.4				P25.1/.../25.8				P26.1				P26.2				P26.3				P27.1/.../27.4				P28.1/.../28.8				P29.1				P29.2			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	- 3				- 2				- 2								- 30				- 30				- 15				- 15				- 12				- 65	± 7	± 7		- 12	± 2	± 2		- 65	± 7	± 7		- 6				- 12				- 65	± 7	± 7		- 12	± 2	± 2	
2 DEAD LOAD (SECONDARY EQUIPMENT)																																																																					
3 OPERATION LOAD 1. FILLING																																																																					
4 OPERATION LOAD 2. PRESSURE																																																																					
5 OPERATION LOAD 3. TEMPERATURE																																																																					
6 OPERATION LOAD 4. DYNAMIC					- 3																																		± 80	± 10	± 30						± 80	± 10	± 30										± 80	± 10	± 30					
7 OPERATION LOAD 5. OTHERS																																																																					
8 TEST LOAD FILLING / EMERGENCY																	- 80				- 80				- 24				- 24				- 29																				- 15				- 29											
9 LIVE LOAD	- 6								- 15																- 30				- 30				- 34																				- 18				- 34											
10 WIND ±X - DIRECTION	-	1	-		-	1	-		-	1	-						-	7.6	-		-	7.6	-		-	12	-		-	12	-		-	1		-	-	2	-		-	2	-		-	2	-		-	-	1	-		-	2	-		-	2	-		-	2	-	
11 WIND ±Y - DIRECTION	-	-	1		-	-	1		-	-	1						-	-	3.1		-	-	3.1		-	-	12		-	-	12		-	-		12	-	-	1		-	1	1		-	1	1		-	-	1		-	-	1		-	-	1		-	-	1	
12 EARTHQUAKE ±X - DIRECTION	-	1	1		-	-	-		-	1	-						-	4	4		-	4	4		-	6	6		-	6	6		-	12		12	-	4	4		-	3	3		-	3	3		-	4	4		-	3	3		-	1	1	
13 EARTHQUAKE ±Y - DIRECTION	-	1	1		-	-	-		-	1	-						-	4	4		-	4	4		-	6	6		-	6	6		-	12		12	-	4	4		-	3	3		-	3	3		-	4	4		-	3	3		-	1	1	
EQUIPMENT - NO.	3310-ANL-002				3310-ANL-002				3310-ANL-002				3220-PKG-304				3220-CHT-003				3220-CHT-006				3230-GCR-001				3230-GCR-002				3220-CHT-004				3220-SCR-002				3220-SCR-002				3220-SCR-002				3220-FDB-001				3220-CHT-005				3220-SCR-003				3220-SCR-003			
DESCRIPTION	PARTICLE				WATER PUMP				WATER TANK				3220-MIL-001				MILL DISCHARGE				MILL SCATS				GRAVITY				GRAVITY				TRASH SCREEN 1				CYCLONE				CYCLONE				CYCLONE				CYCLONE TRASH				TRASH				CYCLONE				CYCLONE			
	SIZE ANALYZER				FOR PSI				FOR PSI				MILL FEED				CHUTE				CHUTE				CONCENTRATOR				CONCENTRATOR				OVERSIZE CHUTE				OVERFLOW				OVERFLOW				OVERFLOW				SCREEN 1				SCREEN 2				OVERFLOW				OVERFLOW			
													CHUTE WINCH												1				2								TRASH SCREEN 1				TRASH SCREEN 1				TRASH SCREEN 1				FEEDBOX				OVERSIZE CHUTE				TRASH SCREEN 2				TRASH SCREEN 2			

LOADING POINT	P29.3				P30.1/.../30.4				P31.1/.../31.8				P32				P33				P34.1/.../34.4				P34.5/.../34.8				P35				P36.1				P36.2				P37.1/.../37.8				P38				P39.1/39.2				P39.3/39.4				P40			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	- 65	± 7	± 7		- 6				- 10								- 45				- 1.3				- 10				- 10								- 15				- 15				- 8				- 4				- 20				- 15							
2 DEAD LOAD (SECONDARY EQUIPMENT)																																																																				
3 OPERATION LOAD 1. FILLING																																	- 60	± 3	± 3		- 50	± 3	± 3												- 30				- 20											
4 OPERATION LOAD 2. PRESSURE																																																																			
5 OPERATION LOAD 3. TEMPERATURE																																																																					
6 OPERATION LOAD 4. DYNAMIC	± 80	± 10	± 30																																																																		
7 OPERATION LOAD 5. OTHERS																																																																					
8 TEST LOAD FILLING / EMERGENCY					- 15				- 70												- 160				- 125																				- 38																				
9 LIVE LOAD					- 18				- 116				- 95				- 1.8																- 125				- 95				- 45				- 7				- 60				- 45											
10 WIND ±X - DIRECTION	-	2	-		-	1	-		-	2	-		-	1	-		-	-	-		-	2	-		-	-	3		-						-	2	-		-	2	-		-	0.6	-		-	-	-		-	2	-		-	-	12	-				
11 WIND ±Y - DIRECTION	-	-	1		-	-	1		-	-	2		-	-	8		-	-	-		-	-	2		-	-	3		-						-	-	1		-	1	1		-	1.2	-		-	-	-		-	0.5	-		-	0.5	-	-	12			
12 EARTHQUAKE ±X - DIRECTION	-	1	1		-	12	12		-	18	18		-	5	5		-	-	-		-	4	4		-	4	4		-						-	7	7		-	5	5		-	6	6		-	-	-		-	3	3		-	3	3	-	-	-	
13 EARTHQUAKE ±Y - DIRECTION	-	1	1		-	12	12		-	18	18		-	5	5		-	-	-		-	4	4		-	4	4		-						-	7	7		-	5	5		-	6	6		-	-	-		-	3	3		-	3	3	-	-	-	
EQUIPMENT - NO.	3220-SCR-003				3220-FDB-002				3220-CHT-001				3220-CNV-001				3220-SMP-003				3220-PKG-002				3230-FDB-002				3230-SCR-002 **				3230-SCR-002 **				3220-DBX-001				3220-SMP-004				3220-SMP-002				3220-SMP-002				3220-CRN-001							
DESCRIPTION	CYCLONE				CYCLONE TRASH				MILL FEED CONVEYOR 2				MILL FEED				CYCLONE				MILL BALL CHARGER				MILL BALL CHARGER				GRAVITY				GRAVITY SCALPING				GRAVITY				CYCLONE				CYCLONE				CYCLONE				CYCLONE				CYCLONE							
	OVERFLOW				SCREEN 1				HEAD CHUTE				CONVEYOR 2				OVERFLOW												SCALPING				SCREEEN 2				SCALPING				OVERFLOW				OVERFLOW				OVERFLOW				OVERFLOW							
	TRASH SCREEN 2				FEEDBOX												SECONDARY												FEED BOX				SCREEEN 2				SCREEEN 2				SCREENING				PARTICLE SIZE				PRIMARY				PRIMARY				CRANE			
																	METALLURGICAL																								DISTRIBUTION				SAMPLER				METALLURGICAL				METALLURGICAL							
																	SAMPLER																								BOX								SAMPLER				SAMPLER							

LOADING POINT	P41.1/.../41.6				P42.1/.../46.8				P43.1/.../43.6				P44				P45.1				P45.2				P46.1/.../46.8				P47				P48				P49				P50				P51*				P52				P53*				P54.1/.../54.5			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	- 4.3				- 15				- 10								- 15				- 15				- 5.7				- 25				- 108				- 40				- 40								- 8								- 11			
2 DEAD LOAD (SECONDARY EQUIPMENT)																																																																			
3 OPERATION LOAD 1. FILLING					- 85												- 60	± 3	± 3		- 50	± 3	± 3						- 147				- 75				- 75																									
4 OPERATION LOAD 2. PRESSURE																																																																		
5 OPERATION LOAD 3. TEMPERATURE																																																																					
6 OPERATION LOAD 4. DYNAMIC																																																																				
7 OPERATION LOAD 5. OTHERS																																																																				
8 TEST LOAD FILLING / EMERGENCY	- 20																								- 33				- 80																														- 20			
9 LIVE LOAD	- 24				- 100				- 20								- 125				- 95				- 39																																		- 24			
10 WIND ±X - DIRECTION	-	10	-		-	2.3	-		-	8	-						-	2	-		-	2	-		-	3	-		-	-	-		-	-	-		-	-	-		-	-	-						-	1	-					
11 WIND ±Y - DIRECTION	-	-	2		-	-	2.3		-	-	8						-	-	1		-	-	1		-	-	5		-	-	-		-	-	-		-	-	-		-	-	-						-	1	-					
12 EARTHQUAKE ±X - DIRECTION	-	12	12		-	5	5		-	6	6						-	6	6		-	5	5		-	12	12		-	4	4		-	6	6		-	4	4		-	4	4						-	1	1					
13 EARTHQUAKE ±Y - DIRECTION	-	12	12		-	5	5		-	6	6						-	6	6		-	5	5		-	12	12		-	4	4		-	6	6		-	4	4		-	4	4						-	1	1					
EQUIPMENT - NO.	3220-FDB-003				3220-CYC-001				3220-PKG-001				3230-FDB-001				3230-SCR-001***				3230-SCR-001***				3220-FDB-004				3220-PKG-003				3220-PKG-301				3220-PKG-302				3220-PKG-303				3220-CHT-002				3310-PMP-502				3220-MIL-001				3230-LND-001			
DESCRIPTION	CYCLONE				MILL CYCLONE CLUSTER				MILL FEED				GRAVITY				GRAVITY SCALPING				GRAVITY				CYCLONE				FIRST FILL AND				3220-MIL-001				3220-MIL-001				3220-MIL-001				MILL ROCK BOX				PARTICLE SIZE				BALL MILL				GRAVITY TAILS			
	OVERFLOW								METALLURGICAL				SCALPING				SCREEEN 1				SCALPING				UNDERFLOW				BACK-UP BALL				MAIN BEARING				GEARBOX				GEARBOX				AND FEED				ANALYZER								LAUNDER 1			
	VELOCITY								SAMPLER				SCREEN 1								SCREEEN 1				VELOCITY				MILL CHARGER				LUBE UNIT				LUBE UNIT				LUBE UNIT				CHUTE				SAMPLE											
	BREAK TANK												FEED BOX												BREAK TANK																												RETURN PUMP											

FOR *
P1.1/../1.3 :REFER DWG. MD-216-3220-EG-GE-GAD-1076 FOR LOAD DETAILS.
P9 :REFER DWG. MD-216-3220-EG-GE-GAD-1048 FOR LOAD DETAILS.
P20 :REFER DWG. MD-216-3220-EG-GE-GAD-1035 FOR LOAD DETAILS
P51 :REFER DWG. MD-216-3220-EG-ME-GAD-1015 FOR LOAD DETAILS
P53 :REFER DWG. MD-216-3220-EG ME-GAD-1003 FOR LOAD DETAILS

NOTE:
** IT ALSO INCLUDE 3230-FDB-002
*** IT ALSO INCLUDE 3230-FDB-001

GENERAL NOTES, LEGEND

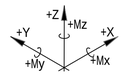
01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm]
OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS



06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

OPERATING PLATFORM- GRATING 5 kN/m² — GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²

HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY,
AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1,6 FOR STRENGTH DESIGN.

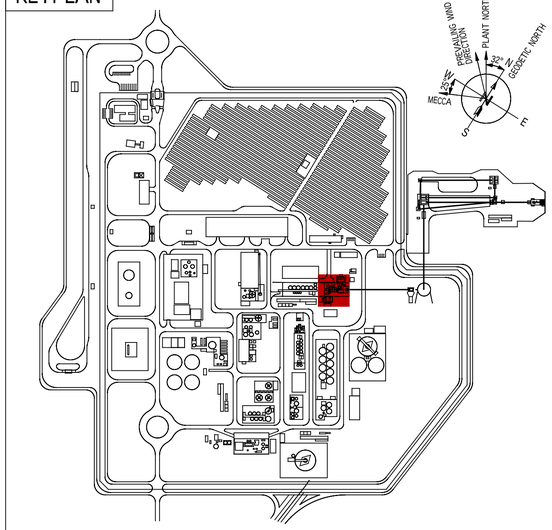
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3220-EG-ME-GAD-1088	LOAD PLAN, PLAN AT EL. (+)931.150								
MD-216-3220-EG-ME-GAD-1087	LOAD PLAN, PLAN AT EL. (+)924.550	A03	17-03-2020	LOAD TABLE UPDATED	MB	TV	SC	RL	
MD-216-3220-EG-ME-GAD-1086	LOAD PLAN, PLAN AT EL. (+)922.150	A02	04-02-2020	ISSUED FOR DESIGN	SF	TV	SC	RL	
MD-216-3220-EG-ME-GAD-1085	LOAD PLAN, PLAN AT EL. (+)913.000	A01	03-01-2020	ISSUED FOR APPROVAL	BR	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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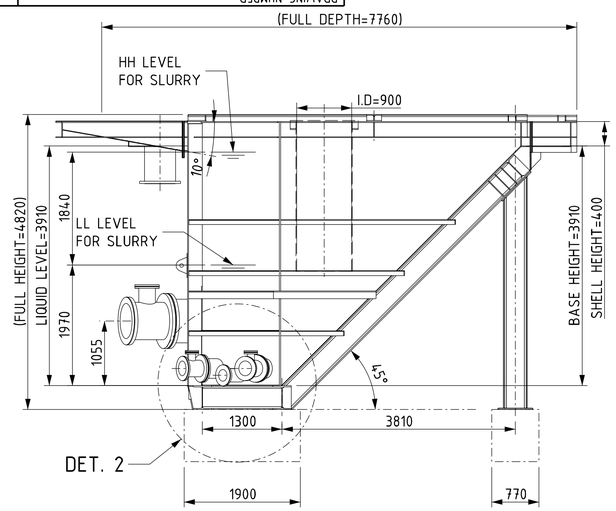
MA'ADEN معادن
Saudi Arabian Mining Company شركة المعادن العربية السعودية

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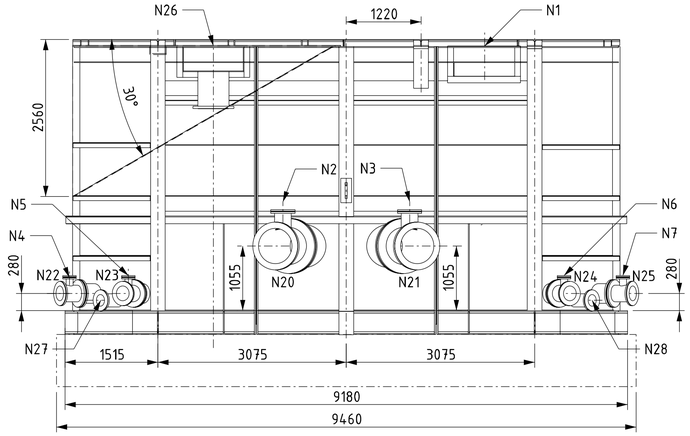
DRAWN	BR	03/01/2020
CHECKED	TV	03/01/2020
ENGINEER	SC	03/01/2020
APPROVED	RL	03/01/2020
EMPLOYER		
SCALE	1:1	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_D_02372_005 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _ZG01 _MLD14 _33205 — DOCUMENT ID: OU100668709 — REVISION: 2

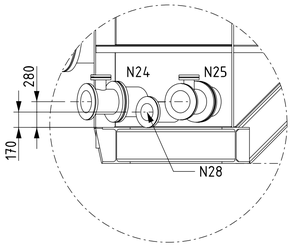
DRAWING NUMBER MD-216-3220-EG-GE-GAD-1048 REV A01



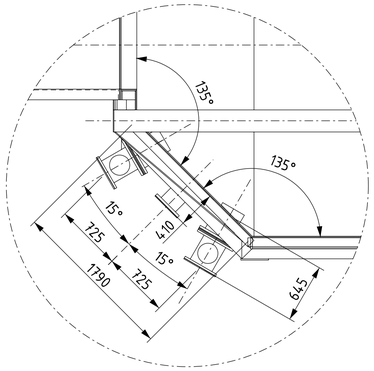
FRONT VIEW



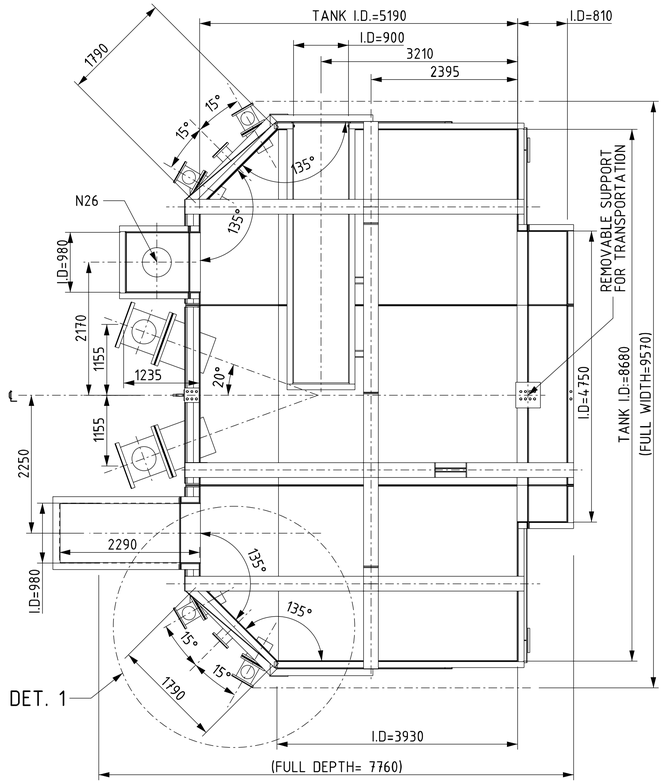
LEFT VIEW



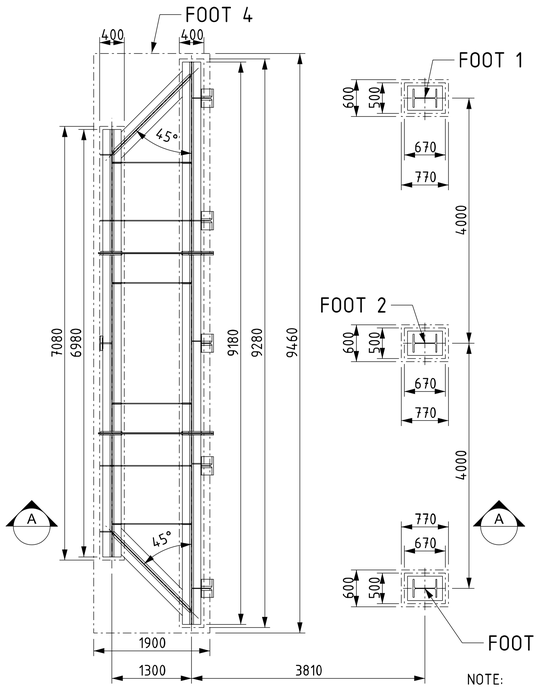
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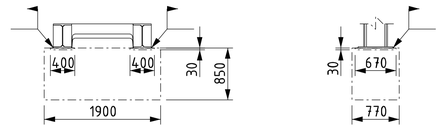
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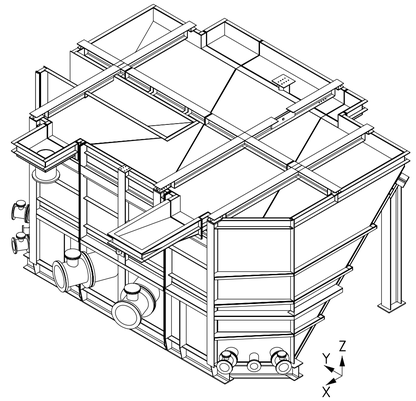
TOP VIEW



ANCHOR PLATES AND FOOTINGS



SECTION A-A



FOOTING COORDINATES

NOTE:
1. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
2. ALL ELEVATION MEASUREMENTS ARE MEASURED FROM BOTTOM SIDE OF BASE PLATE.
3. SHELL NOZZLE ELEVATIONS ARE FROM BOTTOM FROM BOTTOM OF BASE PLATE TO CENTERLINE OF NOZZLE.
4. SHELL NOZZLE PROJECTIONS ARE MEASURED FROM OUTSIDE OF SHELL TO FLANGE FACE.
5. ALL NOZZLE LOCATION, ELEVATION AND ORIENTATION ARE INDICATIVE ONLY AND WILL BE FINALIZED DURING DETAIL DESIGN PHASE.

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3220-TNK-001
DESCRIPTION	BALL MILL DISCHARGE SUMP
CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	0.977 kPa/ SBC 301,2007
CORROSION ALLOWANCE: SHELL INSIDE	2 mm (10 THK. RUBBER LINING PROVIDED)
CORROSION ALLOWANCE: BOTTOM	2 mm (10 THK. RUBBER LINING PROVIDED)
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	4820mmx7720mmx9260mm
SHELL HEIGHT	1385mm
HEIGHT OF BASE	3130mm
CAPACITY: EFFECTIVE VOLUME	100m³
SEISMIC DESIGN /CODE	MD-216-1000-EG-GE-SPC-1001 /SBC 301, 2007

LOAD DATA

TOTAL WEIGHT (FULL)	257900 kg
TOTAL WEIGHT (EMPTY)	35740 kg
RUBBER LINING WEIGHT	1640 kg
TANK WEIGHT	34100 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

LOAD TABLE

MAXIMUM VALUES FOR FOOTING LOADS INLUDING SEISMIC, WIND AND OPERATIONAL LOADS

FOOT	1	2	3	4
MAX SHEAR FXY (kN)	11.9	38.4	13.7	101.7
MAX AXIAL FZ (kN)	61.9	376.1	58.0	1809.8
MIN AXIAL FZ (kN)	22.0	44.6	21.9	261.9
MAX MX (kN/m)	3.5	0.2	3.6	286.0
MAX MY (kN/m)	17.0	57.0	19.6	597.1
MAX MZ (kN/m)	0.1	0.0	0.1	65.9

NOTE! Moments and shear forces can act in any direction

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284/ EN10025-2 S235JR
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A284
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS ±40 ShA

NOZZLES TABLE

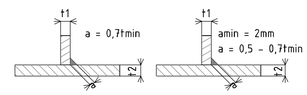
POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N1	-		-	-	-	-	ASME B16.5	SLURRY INLET
N2	DN250	150	-	-	-	-	ASME B16.5	FLUSHING
N3	DN250	150	-	-	-	-	ASME B16.5	FLUSHING
N4	DN100	150	-	-	-	-	ASME B16.5	FLUSHING
N5	DN100	150	-	-	-	-	ASME B16.5	FLUSHING
N6	DN100	150	-	-	-	-	ASME B16.5	FLUSHING
N7	DN100	150	-	-	-	-	ASME B16.5	FLUSHING
N20	DN500	150	-	1235	-	1055	ASME B16.5	OUTLET
N21	DN500	150	-	1235	-	1055	ASME B16.5	OUTLET
N22	DN250	150	-	645	-	280	ASME B16.5	OUTLET(BLINDED FOR FUTURE)
N23	DN250	150	-	645	-	280	ASME B16.5	OUTLET
N24	DN250	150	-	645	-	280	ASME B16.5	OUTLET
N25	DN250	150	-	645	-	280	ASME B16.5	OUTLET
N26	DN600	150	-	-	-	3910	ASME B16.5	EMERGENCY OVERFLOW
N27	DN200	150	-	410	-	170	ASME B16.5	DRAIN
N28	DN200	150	-	410	-	170	ASME B16.5	DRAIN

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	22-11-19	FIRST ISSUE	KL	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER 019045-450-M-G-02315-009 REVISION 0

OUTOTEC DRAWING NUMBER 900707_GAU01_ZG01_MTA01_32209 DOCUMENT ID OU100660901 REVISION 0

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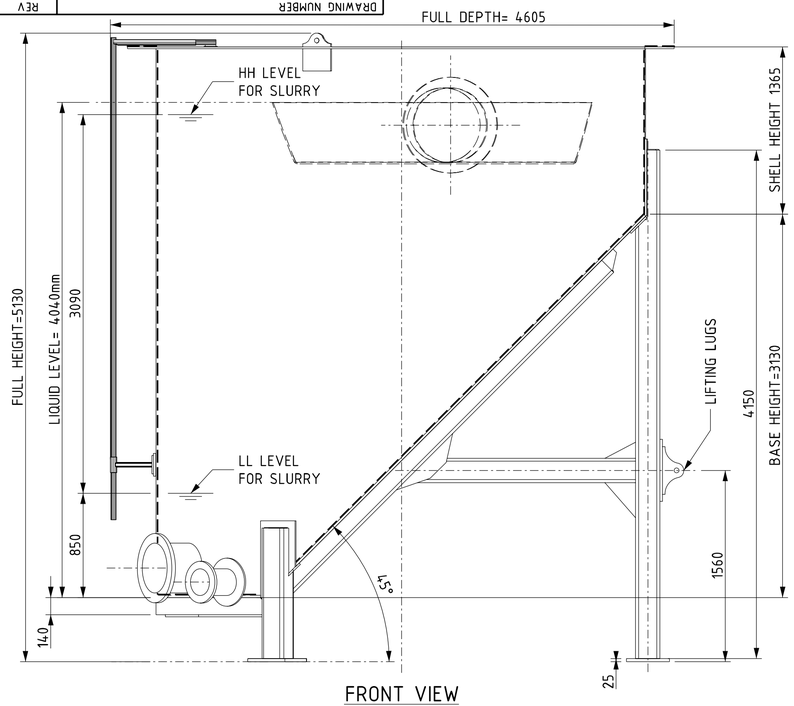
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Saudi Arabian Mining Company شركة التعدين العربية السعودية
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DRAWN	KimLop	22-11-19
CHECKED	TatVie	22-11-19
ENGINEER	SouCha	22-11-19
APPROVED	RicLeh	22-11-19
EMPLOYER		
SCALE 1:50		A1

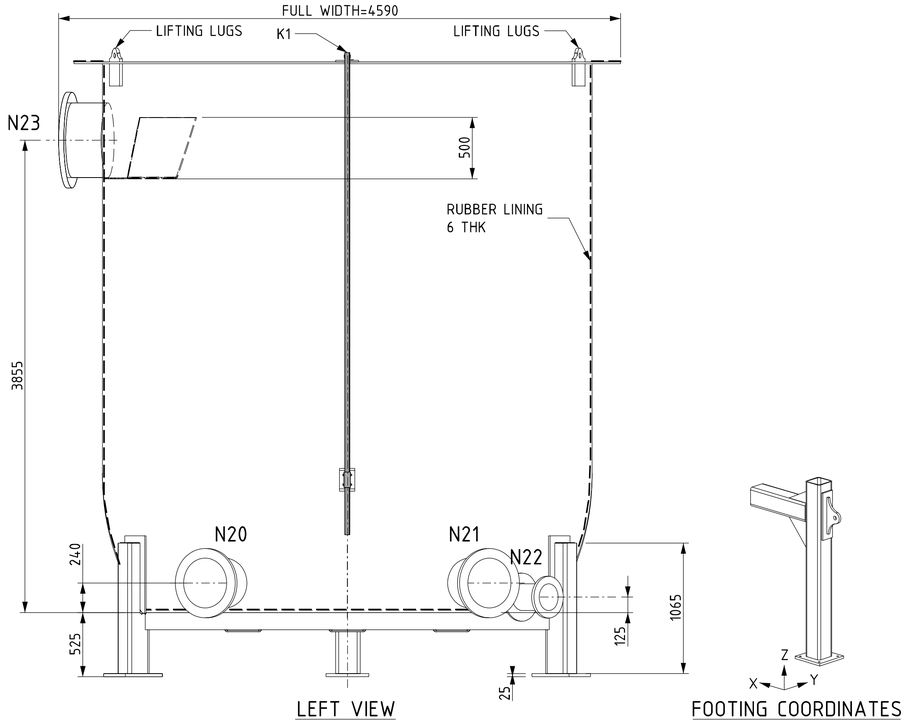
MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
GRINDING
3220-TNK-001 BALL MILL DISCHARGE SUMP

DRAWING NUMBER MD-216-3220-EG-GE-GAD-1048 REV A01

DRAWING NUMBER MD-216-3310-EG-GE-GAD-1076 REV A01

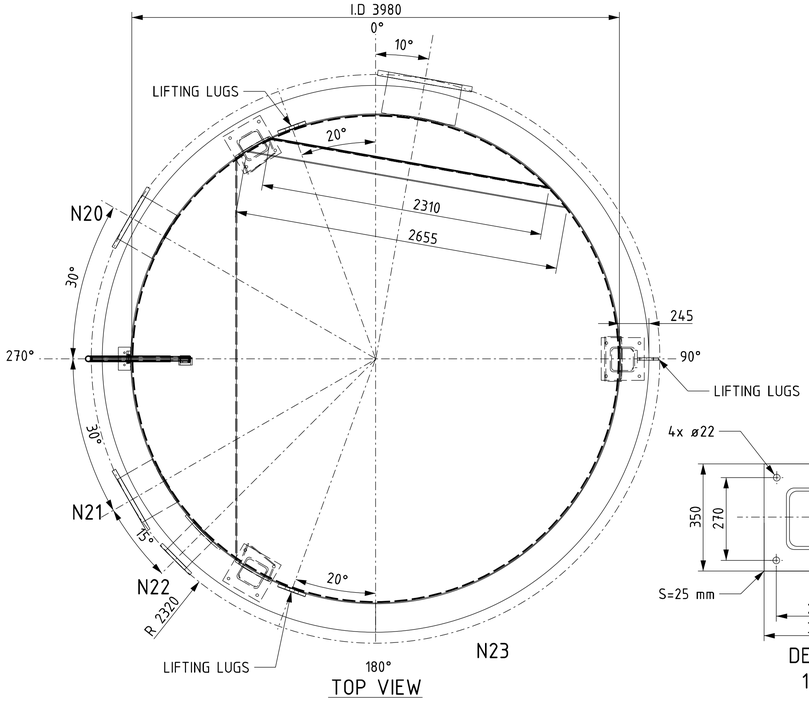


FRONT VIEW

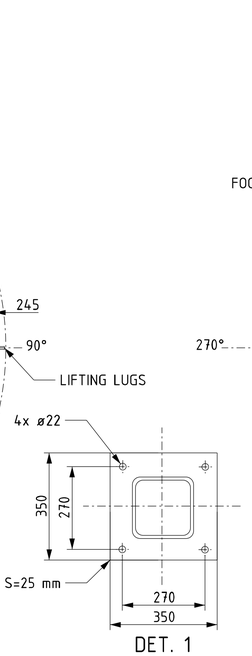


LEFT VIEW

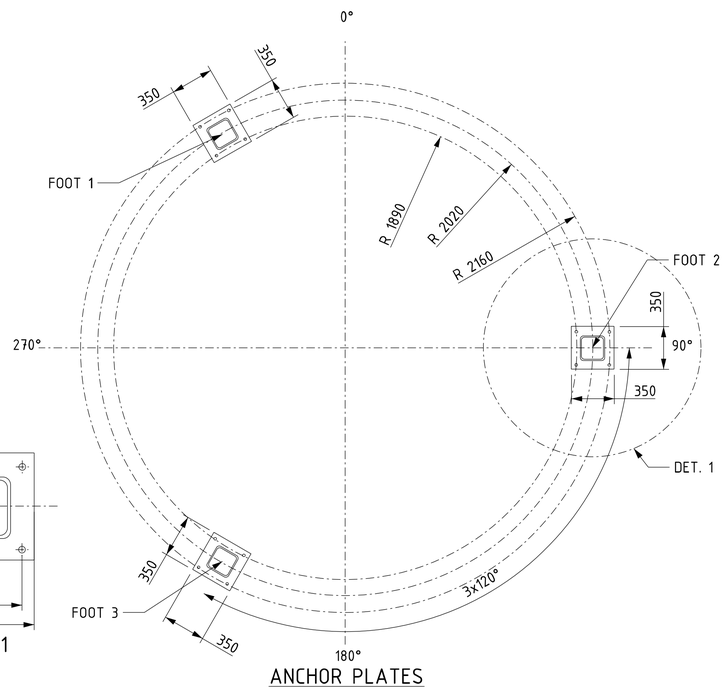
FOOTING COORDINATES



TOP VIEW



DET. 1
1:10



ANCHOR PLATES

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3220-TNK-002
DESCRIPTION	CYCLONE OVERFLOW PUMP SUMP
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	0,977 kPa/ SBC 301,2007
CORROSION ALLOWANCE: SHELL INSIDE	2 mm (6 THK. RUBBER LINING PROVIDED)
CORROSION ALLOWANCE: BOTTOM	2 mm (6 THK. RUBBER LINING PROVIDED)
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	4605mmx4590mmx5130mm
SHELL HEIGHT	1385mm
HEIGHT OF BASE	3130mm
CAPACITY: EFFECTIVE VOLUME	30m³
SEISMIC DESIGN /CODE	MD-216-1000-EG-GE-SPC-1001 /SBC 301, 2007

LOAD DATA

TOTAL WEIGHT (FULL)	58100 kg
TOTAL WEIGHT (EMPTY)	9040 kg
RUBBER LINING WEIGHT	230 kg
TANK WEIGHT	3900 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

LOAD TABLE

MAXIMUM VALUES FOR FOOTING LOADS INLUDING SEISMIC, WIND AND OPERATIONAL LOADS

FOOT	1	2	3
MAX SHEAR FXY (kN)	41.5	57.5	44.2
MAX AXIAL FZ (kN)	226.9	146.1	230.2
MIN AXIAL FZ (kN)	26.7	23.7	21.5
MAX MX (kN/m)	4.3	0.8	7.2
MAX MY (kN/m)	17.1	31.7	17.3
MAX MZ (kN/m)	4.1	0.0	4.3

NOTE! Moments and shear forces can act in any direction

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284/ EN10025-2 S235JR
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A284
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS ±40 ShA

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N20	DN450	150	300°	260	-	240	ASME B16.5	OUTLET
N21	DN450	150	240°	260	-	240	ASME B16.5	OUTLET
N22	DN200	150	225°	260	-	125	ASME B16.5	DRAIN
N23	DN600	150	10°	260	-	3855	ASME B16.5	EMERGENCY OVERFLOW
K1	DN100	150	270°	-	-	-	ASME B16.5	LEVEL INDICATOR/HOLD!

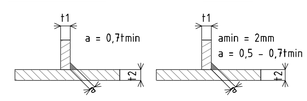
NOTE:
1. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
2. ALL ELEVATION MEASUREMENTS ARE MEASURED FROM BOTTOM SIDE OF BASE PLATE.
3. SHELL NOZZLE ELEVATIONS ARE FROM BOTTOM FROM BOTTOM OF BASE PLATE TO CENTERLINE OF NOZZLE.
4. SHELL NOZZLE PROJECTIONS ARE MEASURED FROM OUTSIDE OF SHELL TO FLANGE FACE.
5. ALL NOZZLE LOCATION, ELEVATION AND ORIENTATION ARE INDICATIVE ONLY AND WILL BE FINALIZED DURING DETAIL DESIGN PHASE.

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	22-11-19	FIRST ISSUE	KL	TV	SC	RL	

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DRAWN	KimLop	22-11-19
CHECKED	TatVie	22-11-19
ENGINEER	SouCha	22-11-19
APPROVED	RicLeh	22-11-19
EMPLOYER		
SCALE	1:25	A1

LARSEN & TOUBRO DRAWING NUMBER 019045 001 X X 000XX 00X — REVISION 0

OUTOTEC DRAWING NUMBER 900707_GAU01_ZG01_MTA01_3222 — DOCUMENT ID OU100666973 — REVISION 0

MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
GRINDING
3220-TNK-002 CYCLONE OVERFLOW PUMP SUMP

DRAWING NUMBER MD-216-3220-EG-GE-GAD-1076 — REV A01

MA'ADEN Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec LARSEN & TOUBRO	
MD-216-3220-EG-CV-CAL-1160	Mansourah Massarah Gold Project	Page	OF
900707_GAU01_ZG01_CED01_32207_OU1 00671085_R2	CALCULATION GRINDING AND CLASSIFICATION-BALL MILL EQUIPMENT FOUNDATION & MISC. FOUNDATION	Date	08.11.2023
O19045_450_C_K_24079_004_R2	AS BUILT	Rev.	D01

ATTACHMENT - 2 : COMMENT RESOLUTION SHEET

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BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-002344, 5 May 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 26 December 2019	Comment by	EPC Response DATE: 02 January 2020	Status	Remarks
			General Comments (cc)				
MD-216-3220-EG-CV-CAL-1160-A01	3220	1	Please improve the title so that in the future people can search or find easily their information request. It should include mention of the equipment name.	Mr. Doron Golan - Bechtel	Equipment name included in the design calculation as per comments.	Closed	